MILITARY SPECIFICATION SHEET

INSULATING COMPONENTS, MOLDED, ELECTRICAL, HEAT SHRINKABLE FLUOROPOLYMER COMPOSITE, SEMI-RIGID, CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the insulating components described herein shall consist of this document and the issue in effect of MIL-I-81765.

REQUIREMENTS:

Continuous operating temperature range: -55°C (-67°F) to + 150°C (+302°F)

Color: Unless otherwise specified, molded components fabricated to the requirements herein shall be furnished in black.

Properties: Properties shall be in accordance with Table I.

TABLE I. Physical properties

Property	Requirement	Test Method (1)
Dimensions, inches As received After shrinkage	Shall be in accordance with the applicable drawing, specification or procurement document	4.6.1 Shrinkage - 200°± 5°C 392°±9°F) for 10 minutes
Elastic memory, %, min Expansion Retraction	 250 90	4.6.2 200°± 5°C (392°± 9°F) for 5 minute; 4 inch stretch
Tensile strength, psi (MPa), min	2200 (15.2)	4.6.3 2 inches/ minute
Tensile Stress at 100% Elongation, psi (MPa) max	1600 (11.0)	4.6.3 2 inches/ minute

AMSC N/A

FSC 5970

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TABLE I (CONTINUED)

Property	Requirement	Test Method
Ultimate elongation, %, min	300	4.6.3 2 inches/ minute
Low temperature flexibility	No cracking	4.6.4 -55°± 1°C (-67°± 2°F) for 4 hours
Heat shock	No dripping, flowing or cracking	4.6.5 225°± 5°C (437°± 9°F) for 4 hours
Heat resistance Tensile strength, psi (MPa), min Elongation, %, min	 1800 (12.4) 250	4.6.6 150°± 3°C (302°± 5°F) for 168 hours
Fluid resistance Tensile strength, psi (MPa), min Elongation, %, min Weight increase, %, max	 1800 (12.4) 250 5	4.6.7 (2)
Corrosion	No corrosion	4.6.8 135°± 2°C (275°± 4°F) for 16 hours
Dielectric strength, volts/mil, (kV/mm), min	200 (7.9)	ASTM D 2671
Volume resistivity, ohm-cm, min	10^{10}	ASTM D 257
Water absorption, %, max	1.0	ASTM D 570 23°± 2°C (73°± 4°F) for 24 hours
Flammability Ave. time of burning, sec., max. Ave. extent of burning, in. (mm) max	 60 1 (25.4)	ASTM D 635
Fungus resistance	Rating of 1 or less	ASTM G 21
Specific gravity, max	1.8	ASTM D 792

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Shelf life (as supplied): The manufacturer shall certify in writing that the molded components, after storage at a temperature of 18° to 35° C (65° to 95° F) for a period of two years, shall conform to the dimensional requirements specified herein and in the contract of purchase order.

NOTE 1: Molded slabs as described in 4.5.2 cannot readily be prepared from this composite material, therefore, test specimens may be prepared directly from a selected molded component of suitable size and geometry.

NOTE 2: Fluid immersion conditions:

Fluid	Time	Temperature
MIL-H-5606	24 hours	50° ± 2°C (122° ± 4°F)
MIL-T-5624 JP-4-Fuel	24 hours	50° ± 2°C (122° ± 4°F)
MIL-L-7808	24 hours	50° ± 2°C (122° ± 4°F)
MIL-L-23699	24 hours	50° ± 2°C (122° ± 4°F)
O-S-1926 5% solution	24 hours	50° ± 2°C (122° ± 4°F)
MIL-A-8243	24 hours	50° ± 2°C (122° ± 4°F)
VV-F-800 Grade DF-2	24 hours	50° ± 2°C (122° ± 4°F)
MIL-D-50030 DS-2	4 hours	25° ± 2°C (77° ± 4°F)

Custodians:
Army - MR
Navy - AS
Air Force - 20

Review Activities:
DLA - GS

Preparing Activity:
Navy - AS
(Project No. 5970-0657-7)

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